To determine the base and ⊥ without explicit labels of sine ($\sin\theta$) and cosine ($\cos\theta$)

1) Identify the angle of intersect (θ), in the triangle on the next page, θ is indicated in the bottom-left part of Δ.

2) The side opposite to the θ is perpendicular if θ is in bottom-left, the vertical side moving upward is typically ⊥. If θ is at bottom right, same as above explained.

3) Then Base & hypotenus will automatically detected.

Remember θ is not right angle.

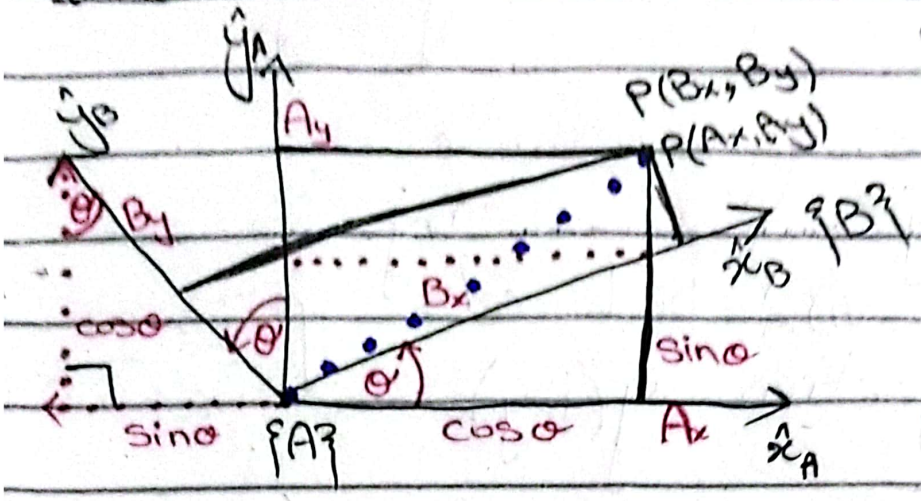
$\sin\theta = \frac{y}{r}$ / opposite / hypotenus

$\cos\theta = \frac{x}{r}$ / adjacent / hypotenus

$\tan\theta = \frac{y}{x}$

$\hookrightarrow \frac{\sin\theta}{\cos\theta} = \frac{y}{r} \cdot \frac{r}{x}$

2D Rotation Matrix about origin at angle θ



∵ Hypotenus is always opposite to right angle.

$$A_P = A_x \hat{x}_A + A_y \hat{y}_A \quad \text{——} (1)$$

$$B_P = B_x \hat{x}_B + B_y \hat{y}_B \quad \text{——} (2)$$

$$\hat{x}_B = \sin\theta \hat{y}_A + \cos\theta \hat{x}_A$$

$$\hat{y}_B = -\sin\theta \hat{x}_A + \cos\theta \hat{y}_A$$

∵ The horizontal component of $\{B\}$ i.e x_B after rotation by θ horizontal component B_x which is $\cos\theta$ and B_y which is $-\sin\theta$

Effect on x and y Projection

The new x-coordinate is primarily influenced by the original $x(B_x)$ but also gets a projection $y(B_y)$.

Since the vertical displacement contributes in the negative x-direction, this introduce -ve sine. $(-B_y \sin\theta)$

B_x contributes +vely to the new-x coordinate through $\cos\theta$

B_y contributes -vely to the new x-coordinate due to downward projection when rotating.

* The sin and cos terms represents the original coordinates onto the new axis defined by the rotation angle θ

* The terms $-B_y \sin\theta$ and $B_x \cos\theta$ reflect the transformation of the original coordinates as the point is rotated, contributing to the opposite axis due to the 90° phase difference in sin and cos

Put $\hat{x}_B$ and $\hat{y}_B$ in (2)

$$B_p = B_x(\sin\theta\,\hat{y}_A + \cos\theta\,\hat{x}_A) + B_y(-\sin\theta\,\hat{x}_A + \cos\theta\,\hat{y}_A)$$

$$B_p = B_x \sin\theta\,\hat{y}_A + B_x \cos\theta\,\hat{x}_A - B_y \sin\theta\,\hat{x}_A + B_y \cos\theta\,\hat{y}_A$$

$$B_p = (B_x \sin\theta + B_y \cos\theta)\,\hat{y}_A + (B_x \cos\theta - B_y \sin\theta)\,\hat{x}_A \quad \text{(3)}$$

$A_p = B_p$

So, Compare (2) & (03)
(Because they share the origin, i.e Starting and ending point is same)

$A_x = B_x \cos\theta - B_y \sin\theta$

$A_y = B_x \sin\theta + B_y \cos\theta$

$$\begin{bmatrix} A_x \\ A_y \end{bmatrix} = \begin{bmatrix} \cos & -\sin \\ \sin & \cos \end{bmatrix} \begin{bmatrix} B_x \\ B_y \end{bmatrix}$$

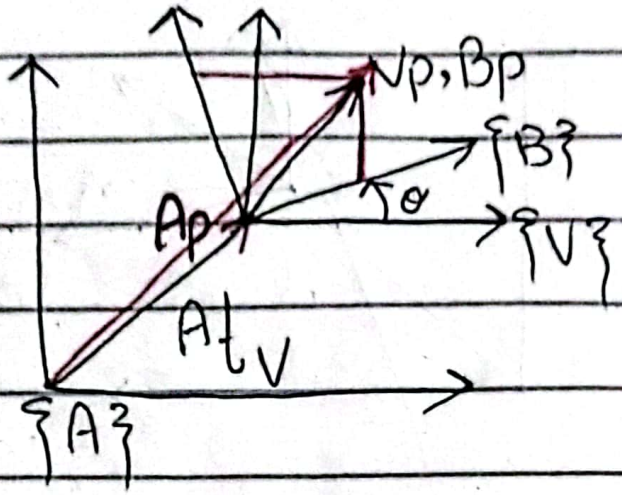
Rotation Matrix about origin at angle θ

$A_p = R^B_p$

$A_p = {}^A R_B \cdot {}^B p$

Transformation Matrix
(Translation + Rotation)

Translation in 2D



$$Vp = {}^{V}R_B \cdot Bp \quad \text{—— ①}$$

$$Ap = {}^{A}t_V + Vp \quad // \{v\} \cdot || \{A\}$$

Put ① to Ap

$$Ap = {}^{A}t_V + {}^{V}R_B \cdot Bp$$

$$Ap = {}^{A}t_B + {}^{A}R_B \cdot Bp \quad // \{v\} || \{A\}$$

$$\begin{bmatrix} A_x \\ A_y \end{bmatrix} = \begin{bmatrix} \cos & -\sin \\ \sin & \cos \end{bmatrix} \begin{bmatrix} B_x \\ B_y \end{bmatrix} + \begin{bmatrix} A_x B \\ A_y B \end{bmatrix}$$

$A_x B = B\cos\theta$

$A_y B = B\sin\theta$

$$\begin{bmatrix} A_x \\ A_y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos & -\sin & A x_B \\ \sin & \cos & A y_B \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ 1 \end{bmatrix}$$

Sign for Homogenous vector

$$\tilde{AP} = \begin{bmatrix} {}^{A}R_B & {}^{A}t_B \\ 0,0 & 1 \end{bmatrix} \tilde{BP}$$

$$\tilde{AP} = {}^{A}t_B \, \tilde{BP}$$

Explanation :-

* Homogenous Transformation Matrix

1) Ap refers to point with respect to A.
2) $^{A}t_V$ refers ~~transformation~~ of Vector $\vec{V}$ with respect to $\vec{A}$.
3) Vp refers to point Rotation Matrix B with respect to A and point with respect to B.

~~$A_x B = A_x \cos\theta - A_y \sin\theta$~~

~~$A_y B = A_x \sin\theta + A_y \cos\theta$~~

- $P = \begin{pmatrix} a \\ b \end{pmatrix}$ $\tilde{P} = \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$

* Why 0 0 1 in Last Row?

The third row [0 0 1] in the matrix makes it a homogenous transformation matrix. Extra row and column are added to allow for translations (and other transformations like scaling). The extra 1 in the last row ensures that the matrix can include both rotation and translation in the same matrix operation (multiplication).

* $A_{tV} = A_{xB}, A_{yB}$

This refers to the new coordinates of vector A after the transformation (rotation and translation) relative to B.

Basically, A_{t_V} is the translation of $\vec{V}$ respect to $\vec{A}$

But, A_{t_V} would represent the displacement/movement of $\vec{V}$ respect to point A.

After applying translation, $\vec{V}$ is moved and new position is A_{t_V}.

A_{t_V} is part of transformation process where you first translate $\vec{V}$ then perform additional operations like rotation or adjusting relative to another vector like Vp.

In summary, A_{t_V} is the translated version of vector $\vec{V}$ relative to point A and determining the final position of point Ap in transformation matrix.

ORTHOGNAL / ORTHONORMAL

$A^T = A^{-1}$

↳ $AA^{-1} = I$

$$AA^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Homogenous Matrix :

CamScanner

$V_i(t) = IR + V_o$

$Q = CV$

$i = C\frac{dV}{dt} \quad \because I = \frac{Q}{t}$

$V_i = RC\frac{dV}{dt} + V_o$

$\frac{1}{RC}.V = \frac{dV}{dt} + V_o.\frac{1}{RC}$

$\frac{1}{RC} \cdot V_{in}(s) = S^* V_{out}(s) + \frac{1}{RC} \cdot V_o(s)$

$\frac{V_{out}}{V_{in}} = \frac{1}{S^*\frac{1}{RC} + 1}$

Robotics

$$V_i(t) = iR + L\frac{di}{dt} + V_o$$

$$i = C\frac{dV_o}{dt}$$

$$Q = CV$$

$$i = \frac{Q}{t}$$

$$V_i = RC\frac{dV_o}{dt} + LC\frac{d^2V_o}{dt^2} + V_o$$

$$\frac{V_i}{LC} = \frac{R}{L}\frac{dV_o}{dt} + \frac{d^2V_o}{dt^2} + \frac{V_o}{LC}$$

derivative in terms of s

$$\frac{V_i(s)}{LC} = \frac{R}{L}S[V_o(s)] + S^2V_o(s) + \frac{V(s)}{LC}$$

$$\frac{V_i(s)}{LC} = V_o(s)\left[\frac{RS}{L} + S^2 + \frac{1}{LC}\right]$$

$$\frac{V_o(s)}{V_i(s)} = \frac{1}{LC} \times \frac{RS}{L} + S^2 + \frac{1}{LC}$$

~~t $\dot{S}$~~

time	S	$\dot{S}$	$\ddot{S}$
0	$\pi/3$	0	0
4	$7\pi/4$	0	0

$$s(t) = q_1(t) = \theta(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$

$$\dot{q}_1(t) = 5At^4 + 4Bt^3 + 3Ct^2 + 2Dt + E$$

$$\ddot{q}_1(t) = 20At^3 + 12Bt^2 + 6Ct + 2D$$

for $t=0$

$$q_1(0) = F \;,\; \dot{q}_1(0) = E \;,\; \ddot{q}_1(0) = 2D$$

for $t=4$

$$q_1(4) = 1024A + 256B + 64C + 16D + 4E + F$$

$$\dot{q}_1(4) = 1280A + 256B + 48C + 8D + E$$

$$\ddot{q}_1(4) = 1280A + 192B + 24C + 2D$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 1024 & 28 & 64 & 16 & 4 & 1 \\ 0 & 0 & 0 & 0 & E & 0 \\ 1280 & 256 & 48 & 8 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1280 & 192 & 24 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} q(0) \\ q(4) \\ \dot{q}(0) \\ \dot{q}(4) \\ \ddot{q}(0) \\ \ddot{q}(4) \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 1024 & 256 & 64 & 16 & 4 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1280 & 256 & 48 & 8 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 1280 & 192 & 24 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} \pi/3 \\ \pi/4 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$T C = Q$$

$$C = T^{-1} Q$$

$$\begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 1024 & 256 & 64 & 16 & 4 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1280 & 256 & 48 & 8 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 1280 & 192 & 24 & 2 & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} \pi/3 \\ \pi/4 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0.0261 \\ -0.2608 \\ 0.6954 \\ 0 \\ 0 \\ 1.0472 \end{bmatrix}$$

$$q(t) = 0.0261t^5 - 0.2608t^4 + 0.6954t^3 + 1.0472$$